DEEPAK SINGH

GenAI Engineer

Agentic Systems & Enterprise RAG | Indore, India (Open to Relocate to Bengaluru) | +91-7566519721 |
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PROFESSIONAL SUMMARY

GenAI Engineer with production experience designing autonomous Agent-to-Agent (A2A) architectures, multi-step LLM reasoning systems, and enterprise-grade Retrieval-Augmented Generation (RAG) platforms. Proven track record architecting stateful LangGraph workflows, function calling routines, and hybrid vector search pipelines processing over 1.5M daily records. Experienced in deploying containerized microservices and CI/CD pipelines across AWS (SageMaker, ECS, Lambda, CodePipeline) with rigorous evaluation (LangSmith, DeepEval) and hallucination mitigation guardrails for enterprise and financial datasets.

TECHNICAL SKILLS

Generative AI & Agentic Systems:	LangChain, LangGraph, Agent-to-Agent (A2A) Collaboration, Model Context Protocol (MCP), ReAct, Function/Tool Calling, OpenAI (GPT-4/GPT-4o), Anthropic Claude, Llama 3, Gemini, LoRA/PEFT.
RAG & Search Technologies:	Hybrid Search (Dense Embeddings + BM25), Cross-Encoder Reranking, Reciprocal Rank Fusion (RRF), Context Windowing, Grounding, Dynamic Citation Systems, Confidence Scoring.
Vector Databases & Storage:	pgvector, FAISS, Pinecone, ChromaDB, Milvus, PostgreSQL, MongoDB, Cassandra.
Cloud, MLOps & CI/CD:	AWS (SageMaker, Bedrock, Lambda, ECS, EC2, S3, CodePipeline), Docker, GitHub Actions, MLflow, LangSmith, Weights & Biases (W&B), DeepEval, Ragas.
Backend & Data Engineering:	Python, FastAPI, Flask, SQL, Snowflake, PySpark, REST APIs, OpenAPI, Server-Sent Events (SSE), Structured & Unstructured ETL Data Pipelines, Role-Based Access Control (RBAC).

PROFESSIONAL EXPERIENCE

BestPeers Infosystem — AI & Data Engineer Indore, India | Sep 2025 – Present

- **Agent-to-Agent (A2A) Orchestration:** Architected autonomous multi-agent pipelines using LangGraph and MCP, coordinating specialized sub-agents for query planning, SQL schema execution, and summary aggregation over 1.5M+ daily records.
- **Hybrid RAG & Hallucination Mitigation:** Deployed hybrid search combining dense vector embeddings (pgvector) and BM25 with cross-encoder reranking, elevating retrieval precision by 25%. Built citation guardrails and token-level confidence scoring to mitigate hallucinations across enterprise queries.

- **Inference CI/CD & Cloud Deployment:** Automated deployment pipelines using Docker and GitHub Actions/CodePipeline to deploy containerized FastAPI inference microservices onto cloud compute, cutting production rollback rates.
- **Evaluation & Observability:** Implemented offline and online LLM evaluation using LangSmith and DeepEval; benchmarked prompts, tracked semantic drift, and reduced tool execution failures across production versions.
- **Throughput & Cost Optimization:** Applied dynamic request batching, prompt pruning, and semantic caching, cutting average end-to-end response latency by 35% and cloud API costs by ~20%.

IBM — AI/ML & Data Engineer Intern Remote | Jul 2024 – Aug 2024

- **Semantic Document Retrieval:** Constructed RAG pipelines leveraging Transformer embeddings, LlamaIndex, and FAISS to query semi-structured and unstructured enterprise documentation.
- **PEFT Model Fine-Tuning:** Applied LoRA and PEFT techniques on open-source LLMs to enforce strict JSON schema outputs, improving downstream task parsing consistency by 25%.
- **Event-Driven Cloud Workflows:** Integrated AWS Lambda and S3 triggers to automate incoming data validation, feature extraction pipelines, and vector index refreshing.

KEY PROJECTS

Ampol Enterprise Financial & Pricing Copilot — Multi-Agent RAG Platform
FastAPI, LangGraph, pgvector, Azure OpenAI, PostgreSQL, Docker, SSE

- Architected a financial decision copilot enabling analysts to query complex pricing models and auditing rules across 1,000+ retail stations.
- Implemented multi-step stateful workflows in LangGraph featuring conversational checkpointing, bounded tool execution, and deterministic guardrails.
- Developed an automated validation layer that cross-checks LLM calculations against authoritative SQL transactional tables, failing closed on numeric discrepancies.
- Streamed real-time responses with grounded citations using Server-Sent Events (SSE) and FastAPI.

AI-Powered Inventory & Demand Forecasting Agentic System
AWS Bedrock, SageMaker, PySpark, LangGraph, Python, REST APIs

- Built an enterprise forecasting platform tracking 6,000+ SKUs by pairing PySpark big data pipelines with AWS Bedrock-powered generative agents.
- Exposed predictive demand insights via modular REST endpoints with human-in-the-loop approval workflows for supply reordering triggers.

EDUCATION & CERTIFICATIONS

Indore Institute of Science and Technology — Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning, CGPA: 7.99 / 10.0 — Indore, India 2021 – 2025

CERTIFICATIONS

Google Certified Professional Machine Learning Engineer • Advanced PySpark & AWS Cloud Architecture